

Notes on Cyclic and Product Groups¹

Every group is a collection of interrelated maximal cycles. Therefore, toward understanding the complex structures of groups, we want to better understand cycles and how they are related by homomorphisms. Here are a few basic results about cycles.

(Abelian Cycles) Cyclic groups are abelian. *Proof.* Let $\mathcal{G}$ be a cyclic group generated by g and consider $a, b \in \mathcal{G}$. Then for some $n, m \in \mathbb{N}$, $ab = g^n g^m = g^{n+m} = g^m g^n = ba$.

Εὐλογητὸς

(The Division Algorithm for Cycles) In a cyclic group $\mathcal{G}$ order n , if $s = qn + r$ with $s, q, r \in \mathbb{Z}$, then $g^s = g^r$ for all $g \in \mathcal{G}$. *Proof.* $g^s = g^{qn+r} = (g^n)^q g^r = g^r$.

ὁ εὐλογήσας

(All Sub-cycles Divisible) If $\mathcal{G}$ is a cyclic group order n and $a \in \mathcal{G}$, then $a^n = e$ and the order of a divides the order of G . *Proof.* Let $\mathcal{G}$ be a cyclic group order n generated by g . Then for any $a \in \mathcal{G}$, there exists an $m \in \mathbb{N}$ such that $a^n = (g^m)^n = g^{mn} = (g^n)^m = e^m = e$. If a is order $k < n$, then $a^k = a^n = e$ and by the division algorithm $k|n$.

ἐξελέξατο

(All Divisible Sub-cycles) If $m|n$, then $C_m < C_n$. *Proof.* Let $mk = n$ for some $k \in \mathbb{N}$ and suppose that g generates C_n . Then, $(g^k)^m = g^{mk} = g^n = e$. Therefore, g^k generates C_m in C_n .

προορίσας

The previous two theorems could be combined into a biconditional characterizing all the cyclic subgroups of a cyclic group, but does this characterize all the subgroups?

(Cyclic Subgroups of Cyclic Groups) Every subgroup of a cyclic group is cyclic. *Proof.* Let $\mathcal{G}$ be a cyclic group order n generated by g and H be a subgroup of $\mathcal{G}$. Consider the smallest $m \in \mathbb{N}$ such that $g^m \in H$. We want to show that g^m generates H . According to the division algorithm, every $h \in H$ can be written in terms of g^m as g^{km+r} where $0 \leq r < m$. We want to show that $r = 0$ and thus every h is a multiple of g^m . Notice that

$$h = (g^m)^k g^r$$

And

$$(g^m)^{-k} h = g^r$$

Therefore, $g^r \in H$. However, since $0 \leq r < m$ and we have supposed m to be the smallest integer such that $g^m \in H$, $r = 0$.

τὴν ἀπολύτρωσιν

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(Unique Cycles) Any two cycles of the same order are isomorphic. *Proof.* Let $\langle g \rangle$ and $\langle h \rangle$ be two cycles with the same order n and consider a map $\phi : \langle g \rangle \rightarrow \langle h \rangle$ such that $g^n \mapsto h^n$ for all $n \in \mathbb{N}$. Being essentially an identity map, ϕ is injective and surjective. Then, $\phi(g^n g^m) = \phi(g^{n+m}) = h^{n+m} = h^n h^m$. Therefore ϕ is a bijective homomorphism and the two groups are isomorphic.

τὴν ἄφεσιν

Now our theorems totally classify the subgroups of cyclic groups. For exploration, we might create subgroup lattice diagrams for all the cyclic groups of order 24 or less. Note that most of these are very simple, especially the prime ordered cycles. We will complete our exploration of the basic results of cycles with a very important homomorphism theorem.

(Folding Cycles) If two elements are in the same cycle, then their homomorphic images are in the same cycle, and the order of that cyclic image divides the order of the original cycle. *Proof.* Let $\mathcal{G}$ be a cyclic group order n generated by g and ϕ be a group homomorphism acting on $\mathcal{G}$. Because $\phi(g^n) = \phi(g)^n$ the images of any elements in the cycle generated by g will be in the same cycle in the image generated by $\phi(g)$. Then, since $g^n = e$, $\phi(g)^n = \phi(e)$. This implies by the division algorithm that $|\phi(g)| \mid |g|$.

ἦς ἐπερίσσευσεν

A great analogy for this theorem is folding a string that forms a closed loop (the mathematical unknot) into two or more overlapping loops. This action retains the order of the original structure while dividing the length of the string evenly. With this theorem we can quickly and confidently account for all the homomorphisms on a group and compute the automorphism group. The group Z_4 , for example, has a limited number of possible homomorphic images. The image must be $\mathbb{Z}_4$, $\mathbb{Z}_2$, or 0. Then, using the fact that homomorphisms map identities to identities and generators to generators, the maps can be pieced together for each image. For exploration, we might find all the homomorphisms and automorphism groups for all the groups of order 8 or less. With this basic understanding of cycles, we can begin to understand how maximal cycles relate. First, we will further explore the direct product with the direct product of cycles in mind.

(Cycle Generators) In a cyclic group $\mathcal{G}$ order n generated by g , the element g^m is also a generator if and only if m and n are relatively prime. *Proof.* Let n be the order of g and suppose that k is the order of g^m . The order $k = n$ if and only if $(g^m)^k = (g^n)^m = e$. But this is true if and only if the least common multiple of m and n is mn . That is, if and only if m and n are relatively prime.

γνώρισας

(Order of Product Elements) The order of an element in a direct product group is the least common multiple of the orders of its components in their respective groups. *Proof.* Let $G \times H$ be a direct product of groups. For each $(a, b) \in G \times H$, $(a, b)^k = (a^k, b^k) = (e, e)$ if and only if $a^k = e$ and $b^k = e$. In this case, k is a multiple of both the order of a in G and the order of b in H . By definition, the least of these common multiples is the order of (a, b) .

ἀνακεφαλαιώσασθαι

(Cyclic Products) The product group $C_m \times C_n$ is isomorphic to C_{mn} if and only if m and n are relatively prime. *Proof.* Suppose that $C_n \times C_m \cong C_{mn}$. Then, there exists a generating element of $C_n \times C_m$ order mn . Since the order of an element in a product group is the least common multiple of the orders of its elements in the component groups, m and n must be relatively prime. On the other hand, if we suppose m and n are relatively prime, the element (a, b) where a generates C_n and b generates C_m has order mn in the product and thus generates it.

ἐκληρώθημεν

(Order of Element Products) For a group $\mathcal{G}$ with $a, b \in \mathcal{G}$, (i) $|a| = |a^{-1}|$, (ii) $|ab| = |ba|$, and (iii) $|a| = |bab^{-1}|$. *Proof.* (i) Since a cycle is abelian the inverse automorphism exists on it and demonstrates that a and a^{-1} generate the same cycle. (ii) Suppose that $|ab| = n$ for some $a, b \in G$. Then, $(ab)^n = e$ and

$$\begin{aligned} (ba)^n &= (ba)(ba) \dots (ba) \\ &= b(ab)(ab) \dots (ab)a \\ &= b(ab)^{n-1}a \\ &= b(ab)^{-1}a \\ &= bb^{-1}a^{-1}a \\ &= e \end{aligned}$$

By this we conclude that $|ab| \leq |ba|$. If we assume rather that $|ba| = m$, by the same argument we can show $|ba| \leq |ab|$. Therefore, $|ab| = |ba|$. (iii) Suppose that $|a| = n$ for some $a \in G$. Since the conjugation maps are automorphisms they preserve the order of the elements.

ἀκούσαντες

(Product and Cycles) The direct product group $G \times H$ is generated by $G \cong G \times \{e\}$ and $H \cong \{e\} \times H$, and the elements of $\mathcal{G}$ commute with the elements of H . *Proof.* In the product group $G \times H$, G and H generate $|G||H|$ distinct elements and thus fill the group. Then, if $(a, e) \in \mathcal{G}$ and $(e, b) \in H$, $(a, e)(e, b) = (a, b) = (e, b)(a, e)$.

πιστεύσαντες

This theorem demonstrates that the direct relation of cycles is an abelian relation.

(Prime Cycles in Abelian Groups) If $\mathcal{G}$ is an abelian group with subgroups C_p and C_q where p and q are relatively prime, then $C_p, C_q < C_{pq} < \mathcal{G}$. *Proof.* Because the order of the intersection of these cyclic subgroups must divide both p and q we know that their intersection is trivial. Let $\mathcal{G}$ be an abelian group order n with subgroups C_p and C_q where p and q are relatively prime. Let a and b be generators of C_p and C_q respectively. Then, $(ab)^{pq} = a^{pq}b^{pq} = (a^p)^q(b^q)^p = e$, and $|ab| |pq|$. But a and b must generate at least $(p-1)(q-1)$ distinct new elements. Therefore, $|ab| = pq$ and $C_{pq} < \mathcal{G}$.

ὁ ἀρραβὼν

We would be wise to suspect that non-abelian groups are not built by abelian relations of cycles. There is indeed a more complex network of relations that we can define by

generalizing the direct product. Instead of combining the two groups in such a way that they can still behave independently so to speak, we can construct products in which movement in one group affects movement in another. That is, for a relation between groups G and H , an action in H induces an automorphism on G (and vice-versa with a change in perspective). We will write this as the semi-direct product $G \rtimes_\phi H$ on the cartesian product $G \times H$ where ϕ is defined as $\phi : H \rightarrow \text{Aut}(G)$ and the group operation is defined as $(g_1, h_1)(g_2, h_2) = (g_1\phi(h_1)(g_2), h_1h_2)$. In this generalization, the direct product is the case in which ϕ is the trivial homomorphism, the action on G being always the non-action. Since there are many such possible homomorphisms we might wonder whether two homomorphism can result in the same product group or if two homeomorphisms can result in two distinct semi-direct products. Both can be true.

(The Semi-Direct Product Group) The semi-direct product is a group. *Proof.* Let G and H be groups. Then,

$$\begin{aligned} ((g_1, h_1)(g_2, h_2))(g_3, h_3) &= (g_1\phi(h_1)(g_2), h_1h_2)(g_3, h_3) \\ &= (g_1\phi(h_1)(g_2)\phi(h_1h_2)(g_3), h_1h_2h_3) \\ &= (g_1\phi(h_1)(g_2)\phi(h_1)\phi(h_2)(g_3), h_1h_2h_3) \\ &= (g_1\phi(h_1)((g_2\phi(h_2)(g_3))), h_1h_2h_3) \\ &= (g_1, h_1)(g_2\phi(h_2)(g_3), h_2h_3) \\ &= (g_1, h_1)((g_2, h_2)(g_3, h_3)) \end{aligned}$$

Note that despite the symmetry, it is important that the automorphism is a function of h_1 and not h_2 . The operation is closed by construction, the image of ϕ being the automorphism group on G . The semi-direct product, like the direct product, has the identity element (e, e) . If $(g, h) \in G \rtimes H$, $(g, h)(e, e) = g\phi(h)(e), he = (g, h)$. And finally, the inverse of (g, h) for a given ϕ is $(\phi(h)^{-1}(g^{-1}), h^{-1})$:

$$(g, h)(\phi(h)^{-1}(g^{-1}), h^{-1}) = (g\phi(h)(\phi(h)^{-1}(g^{-1})), hh^{-1}) = (e, e)$$

εἰς ἔπαινον τῆς δόξης αὐτοῦ

Lastly, before moving on to explore other structures and characteristics of groups, it might be useful to start classifying automorphism groups.

(Automorphisms on Cyclic Groups) The order of the automorphism group on C_n is equal to $\phi(n)$, the Euler Totient Function, and it can be represented as the set of its generators under multiplication. *Proof.* The cyclic group C_n has a set of generators S with a cardinality equal to the Euler Totient Function on n . Let s_0 be one particular generator in S . For all $s \in S$, there exists an automorphism $\tau_s : C_n \rightarrow C_n$ where $s_0 \mapsto s$. Because of the constraint that homomorphisms preserve generators and the fact that $\tau_s(s_0)$ generates the whole image these automorphisms represent all the automorphisms on $\mathbb{Z}_n$. Now, suppose that k and j are generators of $\mathbb{Z}_n$. Then, k and j have no common factors with n , and kj likewise has no common factors with n . So, kj is also a generator. Then, $\tau_k(s_0) = k = s_0^x$ for some $x \in \mathbb{N}$ and $\tau_j(s_0) = j = s_0^y$ for some $y \in \mathbb{N}$. Composing these we find that $(\tau_j\tau_k)(s_0) = ((s_0)^x)^y = (s_0)^{xy} = kj$.

εἰς ἔπαινον τῆς δόξης αὐτοῦ

Based on this analysis, can you determine when automorphism groups are cyclic? What can you discover about the automorphism groups of product groups, direct or otherwise? For exploration, we might create the cycle diagrams for products and look for isomorphisms. What is $\mathbb{Z}_n \rtimes \mathbb{Z}_2$? $V_4 \rtimes \mathbb{Z}_3$? To the contrary, how can the groups we know be split into product representations? Which groups do not seem to be the result of a product? We will explore these more in the next set of notes.